

Electricidad I:

Respuesta completa de un circuito RL a un escalón

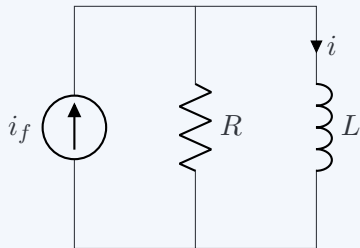
Juan J. Rojas

Instituto Tecnológico de Costa Rica
2 de agosto de 2026

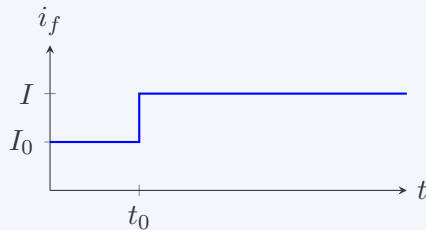
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Circuito RL: respuesta a un escalón



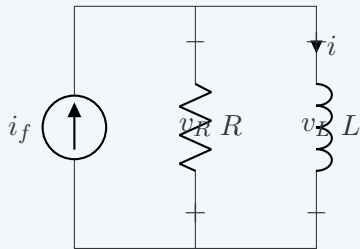
$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$



Se asume que ha pasado mucho tiempo y por lo tanto el inductor tiene una corriente inicial

$$i(t_0) = I_0$$

Circuito RL: respuesta a un escalón

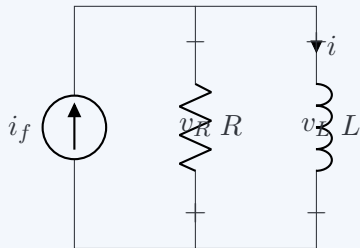


$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

aplicando LVK tenemos que:

$$v_L + v_R = 0$$

Circuito RL: respuesta a un escalón



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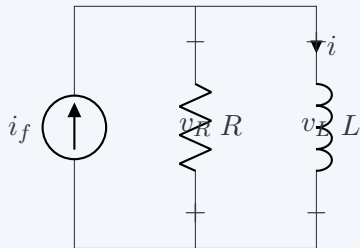
aplicando LVK tenemos que:

$$v_L + v_R = 0$$

y sabemos que para todo $t > t_0$:

$$v_L = L \frac{di}{dt}, v_R = (i - I)R$$

Circuito RL: respuesta a un escalón



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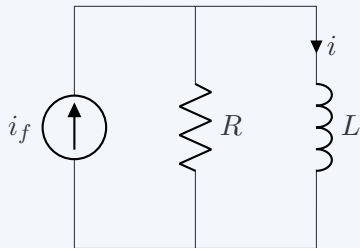
y sabemos que para todo $t > t_0$:

$$v_L = L \frac{di}{dt}, v_R = (i - I)R$$

entonces:

$$L \frac{di}{dt} + (i - I)R = 0$$

Circuito RL: respuesta a un escalón

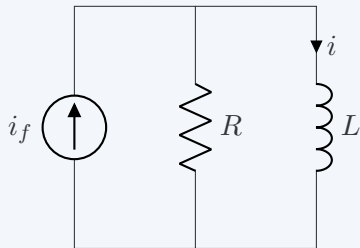


$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

dividiendo entre L :

$$\frac{di}{dt} = -\frac{(i - I)R}{L}$$

Circuito RL: respuesta a un escalón



$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

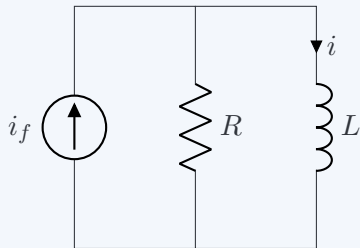
dividiendo entre L :

$$\frac{di}{dt} = -\frac{(i - I)R}{L}$$

reordenando e integrando a ambos lados:

$$\int_{i(t_0)}^{i(t)} \frac{1}{i - I} di = \int_{t_0}^t -\frac{R}{L} dt$$

Circuito RL: respuesta a un escalón

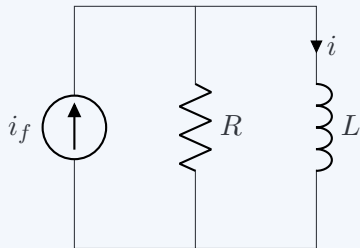


$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

tenemos que:

$$\ln(i - I) \Big|_{i(t_0)}^{i(t)} = -\frac{R}{L} t \Big|_{t_0}^t$$

Circuito RL: respuesta a un escalón



$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

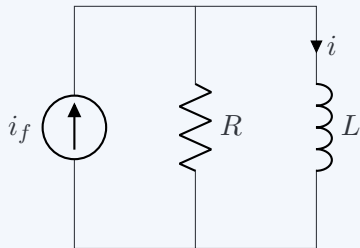
tenemos que:

$$\ln(i - I) \Big|_{i(t_0)}^{i(t)} = -\frac{R}{L} t \Big|_{t_0}^t$$

evaluando:

$$\frac{\ln(i(t) - I)}{\ln(i(t_0) - I)} = -\frac{R}{L}(t - t_0)$$

Circuito RL: respuesta a un escalón

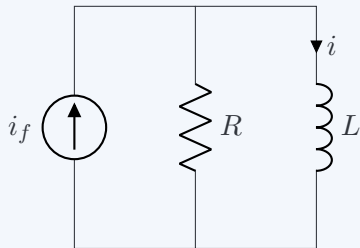


$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

aplicamos la función exponencial a
ambos lados

$$\frac{i(t) - I}{i(t_0) - I} = e^{-(t-t_0)R/L}$$

Circuito RL: respuesta a un escalón



$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

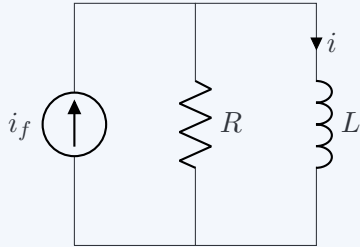
aplicamos la función exponencial a
ambos lados

$$\frac{i(t) - I}{i(t_0) - I} = e^{-(t-t_0)R/L}$$

sea $\tau = L/R$ y $i(t_0) = I_0$, obtenemos:

$$i(t) = I + (I_0 - I)e^{-(t-t_0)/\tau}$$

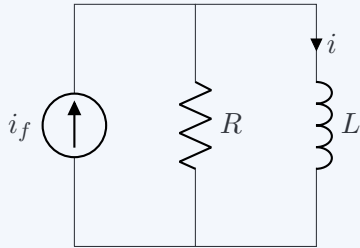
Circuito RL: respuesta a un escalón



$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

$$i(t) = \begin{cases} I_0, & t < t_0 \\ I + (I_0 - I)e^{-(t-t_0)/\tau}, & t \geq t_0 \end{cases}$$

Circuito RL: respuesta a un escalón

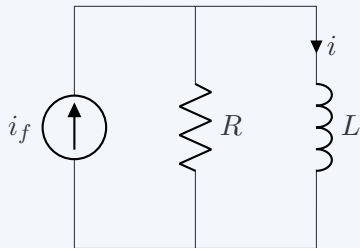


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$$v(t) = L \frac{di}{dt}$$

Circuito RL: respuesta a un escalón



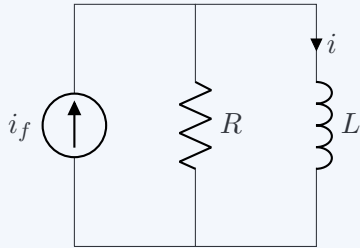
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$$v(t) = L \frac{di}{dt}$$

$$v(t) = \begin{cases} 0, & t < t_0 \\ (I - I_0)R e^{-(t-t_0)/\tau}, & t > t_0 \end{cases}$$

Circuito RL: respuesta a un escalón

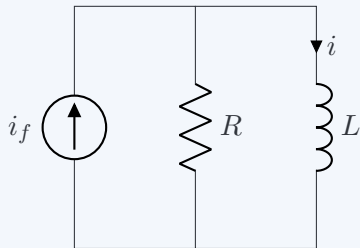


$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

Para $t > t_0$ y reordenando tenemos:

$$i(t) = I_0 e^{-(t-t_0)/\tau} + I(1 - e^{-(t-t_0)/\tau})$$

Circuito RL: respuesta a un escalón

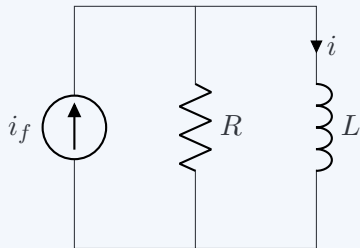


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Para $t > t_0$ y reordenando tenemos:

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Circuito RL: respuesta a un escalón

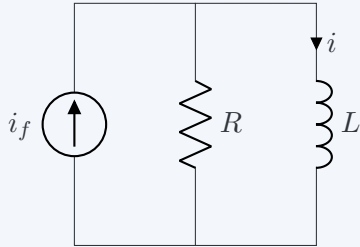


$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

Para $t > t_0$ y reordenando tenemos:

$$i(t) = \underbrace{I_0 e^{-(t-t_0)/\tau}}_{\text{Respuesta natural}} + \underbrace{I(1 - e^{-(t-t_0)/\tau})}_{\text{Respuesta forzada}}$$

Circuito RL: respuesta a un escalón

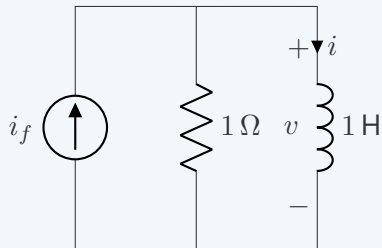


$$i_f(t) = I_0 u(t_0 - t) + I u(t - t_0)$$

Esta respuesta completa puede expresarse como:

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

Ejemplo

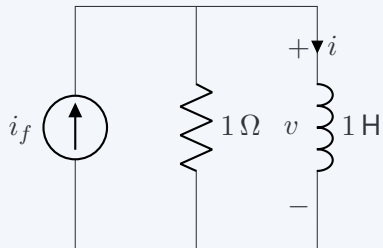


$$i_f(t) = 5\text{ A}[1 + u(t - 1) - u(t - 4)]$$

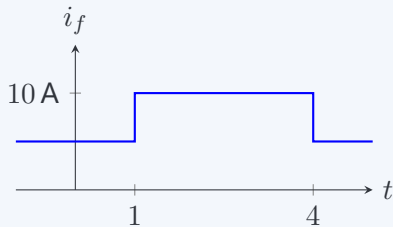
Objetivo:

Averiguar $i(t)$ y $v(t)$ para todo t

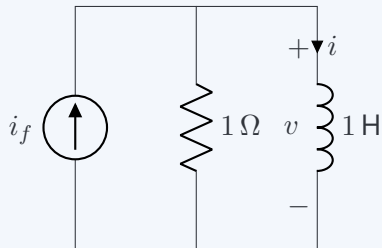
Ejemplo



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Ejemplo

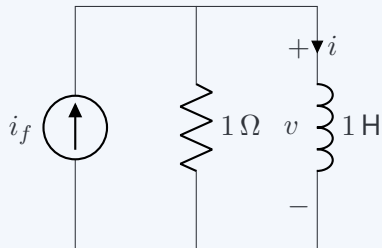


$$i_f(t) = 5\text{ A}[1 + u(t - 1) - u(t - 4)]$$

para $t < 1$:

$$i(t) = 5\text{ A}$$

Ejemplo



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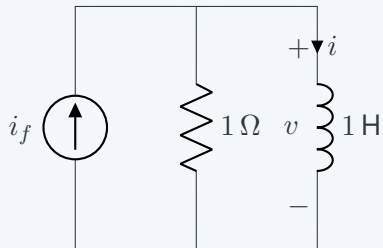
para $t < 1$:

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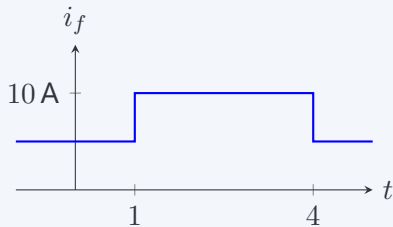
por lo tanto

$$i(1) = 5\text{ A}$$

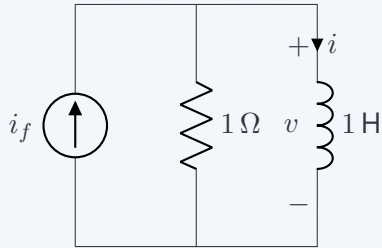
Ejemplo



$$i_f(t) = 5\text{ A}[1 + u(t - 1) - u(t - 4)]$$



Ejemplo

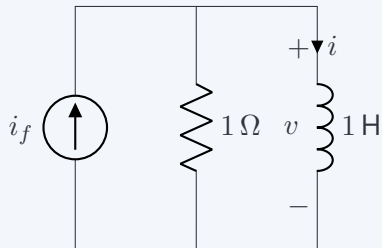


$$i_f(t) = 5\text{ A}[1 + u(t - 1) - u(t - 4)]$$

para $1 \leq t < 4$:

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

Ejemplo



$$i_f(t) = 5 \text{ A} [1 + u(t - 1) - u(t - 4)]$$

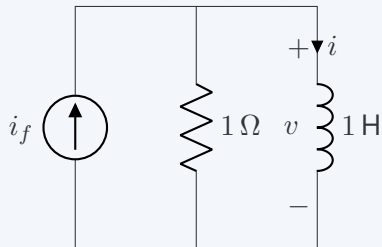
para $1 \leq t < 4$:

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

pero $i(\infty) = 10 \text{ A}$, $\tau = 1 \text{ s}$, $t_0 = 1 \text{ s}$ y
 $i(1) = 5 \text{ A}$, entonces:

$$i(t) = 10 \left(1 - 0,5e^{-(t-1)} \right)$$

Ejemplo



$$i_f(t) = 5 \text{ A} [1 + u(t - 1) - u(t - 4)]$$

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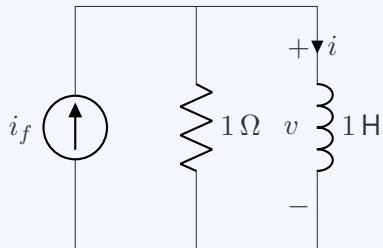
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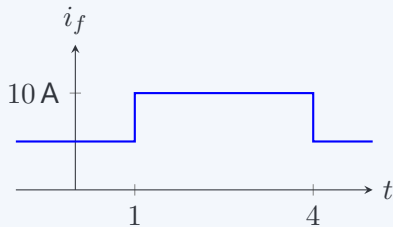
entonces:

$$i(4) = 10 \left(1 - 0,5e^{-3} \right) = 9,751 \text{ A}$$

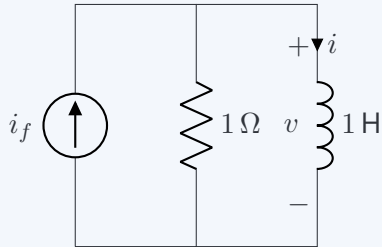
Ejemplo



$$i_f(t) = 5\text{ A}[1 + u(t - 1) - u(t - 4)]$$



Ejemplo

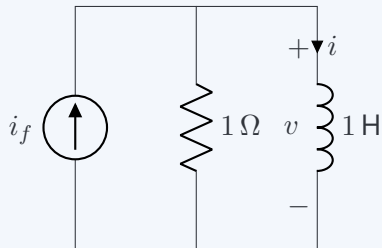


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para $t \geq 4$:

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Ejemplo



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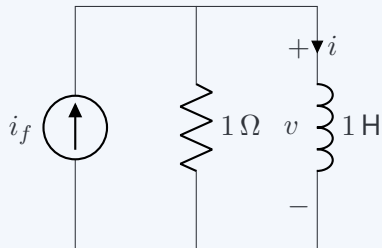
para $t \geq 4$:

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

pero $i(\infty) = 5 \text{ A}$, $\tau = 1 \text{ s}$, $t_0 = 4 \text{ s}$ y
 $i(4) = 9,751 \text{ A}$, entonces:

$$i(t) = 5 + 4,751e^{-(t-4)}$$

Ejemplo

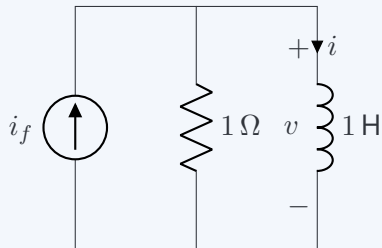


$$i_f(t) = 5\text{ A}[1 + u(t - 1) - u(t - 4)]$$

Finalmente para todo t tenemos que:

$$i(t) = \begin{cases} 5 & \text{A, } t < 1 \\ 10(1 - 0,5e^{-(t-1)}) & \text{A, } 1 \leq t < 4 \\ 5 + 4,751e^{-(t-4)} & \text{A, } t \geq 4 \end{cases}$$

Ejemplo



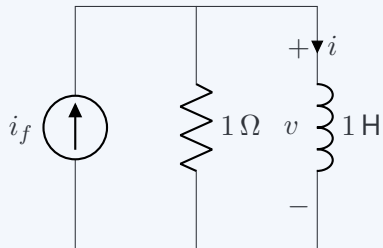
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$$v(t) = \begin{cases} 0 & \text{V, } t < 1 \\ 5e^{-(t-1)} & \text{V, } 1 < t < 4 \\ -4,751e^{-(t-4)} & \text{V, } t > 4 \end{cases}$$

Ejemplo



$$i_f(t) = 5\text{A}[1 + u(t - 1) - u(t - 4)]$$

